

CENTRAL VENOUS CATHETER (CVC) — Procedure One-Pager

US-guided IJ triple-lumen insertion · TeachIM simulation
curriculum · IM resident reference

1 · INDICATIONS

- Vasopressors / periph-incompatible infusion (TPN, vesicants)
- Inadequate peripheral access
- Hemodynamic monitoring (CVP, ScvO₂)
- Hemodialysis (dedicated HD catheter) · transvenous pacing / PA catheter

WRONG line for massive transfusion: TLC lumens are long/narrow → high resistance, slow flow. Use large-bore peripheral IV or introducer/Cordis.

2 · CONTRAINDICATIONS

Hard stops (site)	Thrombus in target vein (non-compressible on US); overlying skin infection; distorted anatomy → pick another site
Cautions (not absolute)	Coagulopathy / low plt (compressible site, IJ>SC); uncooperative pt; severe lung dz (PTX). ESKD & rash are NOT contraindications

Site trade-offs (3SITES, NEJM 2015): IJ = US-guided default. Subclavian = lowest infxn/clot, highest mechanical (PTX). Femoral = fast in a code, highest infxn/clot — avoid elective.

3 · SETUP & STERILE PREP

- **Position:** Trendelenburg to engorge vein.
- **Pre-scan (linear probe):** IJV visualized, predictable path vs carotid, PATENT to compression. Non-compressible = thrombus.
- **TIME-OUT** w/ RN before opening sterile supplies.
- **Maximal barrier:** cap, mask, eye protn, sterile gown+gloves, full-body drape.
- **Prep:** chlorhexidine **scrub x2, 30 s** (iodine = apply & dry).
- **Prepare:** prime guidewire; flush + clamp/cap all lumens.

4 · CANNULATION → CATHETER

- Lidocaine wheal + tract under US.
- **Introducer needle** 45–60°, negative pressure, **FOLLOW-THE-TIP** dynamic US → dark venous blood.
- **Guidewire** >10 cm; **watch telemetry** — withdraw few cm if ectopy.
- Remove needle; **CONFIRM wire in vein, both axes.**
- #11 nick → dilator to vessel depth → catheter over wire (wire out **DISTAL port**, grasp it) to **~12–16 cm** (R IJ, by height).
- Remove wire **AFTER** catheter in; blood return + flush all lumens, Luer-lock; ≥2 sutures + CHG sponge + dressing.

SAFETY: confirm wire IN VEIN before dilating (dilating the carotid = catastrophe). **Never lose the wire** — remove only after catheter placed.

5 · VENOUS vs ARTERIAL

Most reliable = **real-time US of the wire in the vein**. Blood color, easy wire passage, non-pulsatile drip are **UNRELIABLE** in shock/hypoxemia. Unsure → transduce pressure or send a gas before dilating.

6 · CONFIRM & CLEAR

Method	Confirms
CXR	Tip at cavo-atrial junction; no PTX
Bedside US	Catheter in vessel; 10 cc agitated saline → turbulent RA flow <~2 s (tip NOT in RA); anterior lung sliding (PTX sens ~91/spec ~98%)

Place "okay to use" only after tip confirmed at cavo-atrial junction.

MASTERY-LEARNING THRESHOLDS

Stage	Standard
MPS (sim)	Every sim-checklist item performed correctly & independently
UPS (clinical)	Clinical checklist + Global Skills + Entrustment = "without direct supervision"

CLABSI BUNDLE

- Maximal sterile barrier + chlorhexidine prep + checklist in view (Pronovost NEJM 2006)
- CHG-impregnated sponge + sterile dressing
- Avoid femoral for elective lines; remove line when no longer needed